

Basic Stochastic Processes

Brzezniak & Zastawniak

Contents

CHAPTER 1

Martingale Inequalities and Convergence

1. Doob's Martingale Inequalities

Proposition 1.1 (Doob's maximal inequality). *Suppose that $\xi_n, n \in \mathbb{N}$, is a non-negative submartingale (with respect to a filtration $\mathcal{F}_n$). Then for any $\lambda > 0$*

$$\lambda P \left(\max_{k \leq n} \xi_k \geq \lambda \right) \leq E \left(\xi_n 1_{\{\max_{k \leq n} \xi_k \geq \lambda\}} \right),$$

where 1_A is the characteristic function of a set A .

PROOF. We put $\xi_n^* = \max_{k \leq n} \xi_k$ for brevity. For $\lambda > 0$ let us define

$$\tau = \min \{k \leq n : \xi_k \geq \lambda\},$$

if there is a $k \leq n$ such that $\xi_k \geq \lambda$, and $\tau = n$ otherwise. Then τ is a stopping time such that $\tau \leq n$ a.s. Since ξ_n is a submartingale,

$$E(\xi_n) \geq E(\xi_\tau).$$

But

$$E(\xi_\tau) = E(\xi_\tau 1_{\{\xi_n^* \geq \lambda\}}) + E(\xi_\tau 1_{\{\xi_n^* < \lambda\}}).$$

Observe that if $\xi_n^* \geq \lambda$, then $\xi_\tau \geq \lambda$. Moreover, if $\xi_n^* < \lambda$, then $\tau = n$, and so $\xi_\tau = \xi_n$. Therefore

$$E(\xi_n) \geq E(\xi_\tau) \geq \lambda P(\xi_n^* \geq \lambda) + E(\xi_n 1_{\{\xi_n^* < \lambda\}}),$$

It follows that

$$\lambda P(\xi_n^* \geq \lambda) \leq E(\xi_n) - E(\xi_n 1_{\{\xi_n^* < \lambda\}}) = E(\xi_n 1_{\{\xi_n^* \geq \lambda\}}),$$

completing the proof. □

Theorem 1.2 (Doob's maximal L^2 inequality). *If $\xi_n, n \in \mathbb{N}$, is a non-negative square integrable submartingale (with respect to a filtration $\mathcal{F}_n$), then*

$$(4.2) \quad E \left| \max_{k \leq n} \xi_k \right|^2 \leq 4E|\xi_n|^2.$$

PROOF. Put $\xi_n^* = \max_{k \leq n} \xi_k$. By Exercise 1.9, Proposition 1.1, the Fubini theorem and finally the Cauchy-Schwarz inequality

$$\begin{aligned} E|\xi_n^*|^2 &= 2 \int_0^\infty t P(\xi_n^* > t) dt \leq 2 \int_0^\infty E(\xi_n 1_{\{\xi_n^* \geq t\}}) dt \\ &= 2 \int_0^\infty \left(\int_{\{\xi_n^* \geq t\}} \xi_n dP \right) dt = 2 \int_\Omega \xi_n \left(\int_0^{\xi_n^*} dt \right) dP \\ &= 2 \int_\Omega \xi_n \xi_n^* dP = 2E(\xi_n \xi_n^*) \leq 2(E|\xi_n|^2)^{1/2} (E|\xi_n^*|^2)^{1/2}. \end{aligned}$$

Dividing by $(E|\xi_n|^2)^{1/2}$, we get (4.2). □

The proof of Doob's Convergence Theorem in the next section hinges on an inequality involving the number of upcrossings.

DEFINITION 1.3. *Given an adapted sequence of random variables $\xi_1, \xi_2, \dots$ and two real numbers $a < b$, we define a gambling strategy $\alpha_1, \alpha_2, \dots$ by putting*

$$\alpha_1 = 0$$

and for $n = 1, 2, \dots$

$$\alpha_{n+1} = \begin{cases} 1 & \text{if } \alpha_n = 0 \text{ and } \xi_n < a, \\ 1 & \text{if } \alpha_n = 1 \text{ and } \xi_n \leq b, \\ 0 & \text{otherwise.} \end{cases}$$

*It will be called the **upcrossings strategy**. Each $k = 1, 2, \dots$ such that $\alpha_k = 1$ and $\alpha_{k+1} = 0$ will be called an upcrossing of the interval $[a, b]$. The upcrossings form a (finite or infinite) increasing sequence*

$$u_1 < u_2 < \dots$$

The number of upcrossings made up to time n , that is, the largest k such that $u_k \leq n$ will be denoted by $U_n[a, b]$ (we put $U_n[a, b] = 0$ if no such k exists).

The meaning of the above definition is this. Initially, we refrain from playing the game and wait until ξ_n becomes less than a . As soon as this happens, we start playing unit stakes at each round of the game and continue until ξ_n becomes greater than b . At this stage we refrain from playing again, wait until ξ_n becomes less than a , and so on. The strategy α_n is defined in such a way that $\alpha_n = 0$ whenever we refrain from playing the n -th game, and $\alpha_n = 1$ otherwise. During each run of consecutive games with $\alpha_n = 1$ the process ξ_n crosses the interval $[a, b]$, starting below a and finishing above b . This is what is meant by an upcrossing. Observe that each upcrossing will increase our total winnings by at least $b - a$. For convenience, we identify each upcrossing with its last step k , such that $\alpha_k = 1$ and $\alpha_{k+1} = 0$. A typical sample path of the upcrossings strategy is shown in Figure 1.

FIGURE 1. Typical paths of ξ_n , α_n and ζ_n ; upcrossings are indicated by bold lines

EXERCISE 1.1. *Verify that the upcrossings strategy α_n is indeed a gambling strategy.*

Hint: You want to prove that α_n is $\mathcal{F}_{n-1}$ -measurable for each n . Since the upcrossings strategy is defined by induction, a proof by induction on n may be your best bet.

SOLUTION 1.1. *Because $\alpha_1 = 0$ is constant, it is $\mathcal{F}_0 = \{\emptyset, \Omega\}$ -measurable. Suppose that α_n is $\mathcal{F}_{n-1}$ -measurable for some $n = 1, 2, \dots$. Then*

$$\{\alpha_n = 0, \xi_n < a\} \cup \{\alpha_n = 1, \xi_n \leq b\} \in \mathcal{F}_n$$

because $\mathcal{F}_{n-1} \subset \mathcal{F}_n$ and ξ_n is $\mathcal{F}_n$ -measurable. This means that

$$\alpha_{n+1} = 1_{\{\alpha_n=0, \xi_n < a\} \cup \{\alpha_n=1, \xi_n \leq b\}}$$

is $\mathcal{F}_n$ -measurable. By induction it follows that α_n is $\mathcal{F}_{n-1}$ -measurable for each $n = 1, 2, \dots$, so $\alpha_1, \alpha_2, \dots$ is a gambling strategy.

Lemma 1.4 (Upcrossings Inequality). *If $\xi_1, \xi_2, \dots$ is a supermartingale and $a < b$, then*

$$(b - a)E(U_n[a, b]) \leq E((\xi_n - a)^-).$$

By x^- we denote the negative part of a real number x , i.e. $x^- = \max\{0, -x\}$.

PROOF. Let

$$\zeta_n = \alpha_1(\xi_1 - \xi_0) + \cdots + \alpha_n(\xi_n - \xi_{n-1})$$

be the total winnings at step $n = 1, 2, \dots$ if the upcrossings strategy is followed, see Figure 1. It will be convenient to put $\zeta_0 = 0$. By Proposition 3.1 (one cannot beat the system using a gambling strategy) ζ_n is a supermartingale.

Let us fix an n and put $k = U_n[a, b]$, so that

$$0 < u_1 < u_2 < \cdots < u_k \leq n.$$

Clearly, each upcrossing increases the total winnings by $b - a$,

$$\zeta_{u_i} - \zeta_{u_{i-1}} \geq b - a$$

for $i = 1, \dots, k$. (We put $u_0 = 0$ for simplicity.) Moreover,

$$\zeta_n - \zeta_{u_k} \geq -(\xi_n - a)^-.$$

It follows that

$$\zeta_n \geq (b - a)U_n[a, b] - (\xi_n - a)^-.$$

Taking the expectation on both sides, we get

$$E(\zeta_n) \geq (b - a)E(U_n[a, b]) - E((\xi_n - a)^-).$$

But ζ_n is a supermartingale, so

$$0 = E(\zeta_1) \geq E(\zeta_n),$$

which proves the Upcrossings Inequality. $\square$

2. Doob's Martingale Convergence Theorem

Theorem 1.5 (Doob's Martingale Convergence Theorem). Suppose that $\xi_1, \xi_2, \dots$ is a supermartingale (with respect to a filtration $\mathcal{F}_1, \mathcal{F}_2, \dots$) such that

$$\sup_n E(|\xi_n|) < \infty.$$

Then there is an integrable random variable ξ such that

$$\lim_{n \rightarrow \infty} \xi_n = \xi \quad \text{a.s.}$$

REMARK 1.1. In particular, the theorem is valid for martingales because every martingale is a supermartingale. It is also valid for submartingales, since ξ_n is a submartingale if and only if $-\xi_n$ is a supermartingale.

REMARK 1.2. Observe that even though all the ξ_n as well as the limit ξ are integrable random variables, it is claimed only that ξ_n trends to ξ a.s. Note that no convergence in L^1 is asserted.

PROOF (Proof of Doob's Martingale Convergence Theorem). By the Upcrossings Inequality (Lemma 1.4)

$$E(U_n[a, b]) \leq \frac{E((\xi_n - a)^-)}{b - a} \leq \frac{M + |a|}{b - a} < \infty,$$

where

$$M = \sup_n E(|\xi_n|) < \infty.$$

Since $U_n[a, b]$ is a non-decreasing sequence, it follows that

$$E\left(\lim_{n \rightarrow \infty} U_n[a, b]\right) = \lim_{n \rightarrow \infty} E(U_n[a, b]) \leq \frac{M + |a|}{b - a} < \infty.$$

This implies that

$$P\left\{\lim_{n \rightarrow \infty} U_n[a, b] < \infty\right\} = 1$$

for any $a < b$. Since the set of all pairs of rational numbers $a < b$ is countable, the event

$$(4.3) \quad A = \bigcap_{a < b \text{ rational}} \left\{\lim_{n \rightarrow \infty} U_n[a, b] < \infty\right\}$$

has probability 1. (The intersection of countably many events has probability 1 if each of these events has probability 1.)

We claim that the sequence ξ_n converges a.s. to a limit ξ . Consider the set

$$B = \left\{\liminf_n \xi_n < \limsup_n \xi_n\right\} \subset \Omega$$

on which the sequence ξ_n fails to converge. Then for any $\omega \in B$ there are rational numbers a, b such that

$$\liminf_n \xi_n(\omega) < a < b < \limsup_n \xi_n(\omega),$$

implying that $\lim_{n \rightarrow \infty} U_n[a, b](\omega) = \infty$. This means that B and the event A in (4.3) are disjoint, so $P(B) = 0$, since $P(A) = 1$, which proves the claim.

It remains to show that the limit ξ is an integrable random variable. By Fatou's lemma

$$\begin{aligned} E(|\xi|) &= E(\liminf_n |\xi_n|) \\ &\leq \liminf_n E(|\xi_n|) \\ &< \sup_n E(|\xi_n|) < \infty. \end{aligned}$$

This completes the proof. □

EXERCISE 1.2. Show that if ξ_n is a non-negative supermartingale, then it converges a.s. to an integrable random variable.

Hint: To apply Doob's Theorem all you need to verify is that the sequence ξ_n is bounded in L^1 , i.e. the supremum of $E(|\xi_n|)$ is less than ∞ .

SOLUTION 1.2. For a non-negative supermartingale

$$\sup_n E(|\xi_n|) = \sup_n E(\xi_n) \leq E(\xi_1) = E(|\xi_1|) < \infty,$$

since

$$E(\xi_n) \leq E(\xi_1)$$

for each $n = 1, 2, \dots$. Thus Doob's Martingale Convergence Theorem implies that ξ_n converges a.s. to an integrable limit.

3. Uniform Integrability and L^1 Convergence of Martingales

The conditions of Doob's theorem imply pointwise (a.s.) convergence of martingales. In this section we shall study convergence in L^1 . To this end we introduce a stronger condition called uniform integrability. Proposition 1.7 shows that it is a necessary condition for L^1 convergence. In Theorem 1.9 we prove that uniform integrability is in fact sufficient for a martingale to converge in L^1 . This enables us to show that each integrable martingale is of the form $E(\xi|\mathcal{F}_n)$. As an application we give a martingale proof of Kolmogorov's 0-1 law.

EXERCISE 1.3. *Show that a random variable ξ is integrable if and only if for every $\varepsilon > 0$ there exists an $M > 0$ such that*

$$\int_{\{|\xi| > M\}} |\xi| dP < \varepsilon.$$

Hint: Split Ω into two sets: $\{|\xi| > M\}$ and $\{|\xi| \leq M\}$. The integrals of $|\xi|$ over these sets must add up to $E(|\xi|)$. As M increases, one of the integrals increases, while the other one decreases. Investigate their limits as $M \rightarrow \infty$.

SOLUTION 1.3. Necessity. Suppose that ξ is integrable. It follows that

$$P\{|\xi| < \infty\} = 1.$$

The sequence of random variables $|\xi|1_{\{|\xi| > M\}}$ indexed by $M = 1, 2, \dots$ is monotone and

$$|\xi|1_{\{|\xi| > M\}} \searrow 0 \quad \text{as } M \rightarrow \infty$$

on the set $\{|\xi| < \infty\}$, i.e. a.s. By the monotone convergence theorem for integrals

$$\int_{\{|\xi| > M\}} |\xi| dP \searrow 0 \quad \text{as } M \rightarrow \infty.$$

It follows that for every $\varepsilon > 0$ there exists an $M > 0$ such that

$$\int_{\{|\xi| > M\}} |\xi| dP < \varepsilon.$$

Sufficiency. Take $\varepsilon = 1$. There exists an $M > 0$ such that

$$\int_{\{|\xi| > M\}} |\xi| dP < 1.$$

Then

$$\begin{aligned} E(|\xi|) &= \int_{\Omega} |\xi| dP \\ &= \int_{\{|\xi| > M\}} |\xi| dP + \int_{\{|\xi| \leq M\}} |\xi| dP \\ &< 1 + MP\{|\xi| \leq M\} \\ &\leq 1 + M < \infty. \end{aligned}$$

Thus, for any sequence ξ_n of integrable random variables and any $\varepsilon > 0$ there is a sequence of numbers $M_n > 0$ such that

$$\int_{\{|\xi_n| > M_n\}} |\xi_n| dP < \varepsilon.$$

If the M_n are independent of n , then we say that the sequence ξ_n is uniformly integrable.

DEFINITION 1.6. A sequence $\xi_1, \xi_2, \dots$ of random variables is called **uniformly integrable** if for every $\varepsilon > 0$ there exists an $M > 0$ such that

$$\int_{\{|\xi_n| > M\}} |\xi_n| dP < \varepsilon$$

for all $n = 1, 2, \dots$

EXERCISE 1.4. Let $\Omega = [0, 1]$ with the σ -field of Borel sets and Lebesgue measure. Take

$$\xi_n = n1_{(0, \frac{1}{n})}.$$

Show that the sequence $\xi_1, \xi_2, \dots$ is not uniformly integrable.

Hint: What is the integral of ξ_n over $\{\xi_n > M\}$ if $n > M$?

SOLUTION 1.4. For any $M > 0$ and any $n > M$ we have

$$(0, \frac{1}{n}) = \{\xi_n > M\},$$

so

$$\int_{\{\xi_n > M\}} \xi_n dP = \int_{(0, \frac{1}{n})} n dP = 1.$$

This means that there is no $M > 0$ such that for all n

$$\int_{\{\xi_n > M\}} \xi_n dP < \frac{1}{2}.$$

The sequence ξ_n is not uniformly integrable.

Proposition 1.7. Uniform integrability is a necessary condition for a sequence $\xi_1, \xi_2, \dots$ of integrable random variables to converge in L^1 .

Lemma 1.8. If ξ is integrable, then for every $\varepsilon > 0$ there is a $\delta > 0$ such that

$$P(A) < \delta \implies \int_A |\xi| dP < \varepsilon.$$

PROOF (Proof of Lemma 1.8). Let $\varepsilon > 0$. Since ξ is integrable, by Exercise 1.3 there is an $M > 0$ such that

$$\int_{\{|\xi| > M\}} |\xi| dP < \frac{\varepsilon}{2}.$$

Now

$$\begin{aligned} \int_A |\xi| dP &= \int_{A \cap \{|\xi| \leq M\}} |\xi| dP + \int_{A \cap \{|\xi| > M\}} |\xi| dP \\ &\leq \int_A M dP + \int_{\{|\xi| > M\}} |\xi| dP \\ &< MP(A) + \frac{\varepsilon}{2}. \end{aligned}$$

Let $\delta = \frac{\varepsilon}{2M}$. Then

$$P(A) < \delta \implies \int_A |\xi| dP < \varepsilon,$$

as required. □

EXERCISE 1.5. Let ξ be an integrable random variable and $\mathcal{F}_1, \mathcal{F}_2, \dots$ a filtration. Show that $E(\xi | \mathcal{F}_n)$ is a uniformly integrable martingale.

Hint: Use Lemma 1.8.

SOLUTION 1.5. In Example 3.4 it was verified that $\xi_n = E(\xi|\mathcal{F}_n)$ is a martingale. Let $\varepsilon > 0$. By Lemma 1.8 there is a $\delta > 0$ such that

$$P(A) < \delta \implies \int_A |\xi| dP < \varepsilon.$$

By Jensen's inequality $|\xi_n| \leq E(|\xi||\mathcal{F}_n)$ a.s., so

$$E(|\xi|) \geq E(|\xi_n|) \geq \int_{\{|\xi_n| \geq M\}} |\xi_n| dP \geq MP\{|\xi_n| > M\}.$$

If we take $M > E(|\xi|)/\delta$, then

$$P\{|\xi_n| > M\} < \delta.$$

Since $\{|\xi_n| > M\} \in \mathcal{F}_n$, it follows that

$$\int_{\{|\xi_n| > M\}} |\xi_n| dP \leq \int_{\{|\xi_n| > M\}} E(|\xi||\mathcal{F}_n) dP = \int_{\{|\xi_n| > M\}} |\xi| dP < \varepsilon,$$

proving that $\xi_n = E(\xi|\mathcal{F}_n)$ is a uniformly integrable sequence.

PROOF (Proof of Proposition 1.7). Suppose that $\xi_n \rightarrow \xi$ in L^1 , i.e. $E|\xi_n - \xi| \rightarrow 0$. We take any $\varepsilon > 0$. There is an integer N such that

$$n \geq N \implies E|\xi_n - \xi| < \frac{\varepsilon}{2}.$$

By Lemma 1.8 there is a $\delta > 0$ such that

$$P(A) < \delta \implies \int_A |\xi| dP < \frac{\varepsilon}{2}.$$

Taking a smaller $\delta > 0$ if necessary, we also have

$$P(A) < \delta \implies \int_A |\xi_n| dP < \varepsilon \quad \text{for } n = 1, \dots, N.$$

We claim that there is an $M > 0$ such that

$$P\{|\xi_n| > M\} < \delta$$

for all n . Indeed, since

$$E(|\xi_n|) \geq \int_{\{|\xi_n| > M\}} |\xi_n| dP \geq MP\{|\xi_n| > M\},$$

it suffices to take

$$M = \frac{1}{\delta} \sup_n E(|\xi_n|).$$

(Because the sequence ξ_n converges in L^1 , it is bounded in L^1 , so the supremum is $< \infty$.)

Now, since $P\{|\xi_n| > M\} < \delta$,

$$\begin{aligned} \int_{\{|\xi_n| > M\}} |\xi_n| dP &\leq \int_{\{|\xi_n| > M\}} |\xi| dP + \int_{\{|\xi_n| > M\}} |\xi_n - \xi| dP \\ &\leq \int_{\{|\xi_n| > M\}} |\xi| dP + E(|\xi_n - \xi|) \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

for any $n > N$ and

$$\int_{\{|\xi_n| > M\}} |\xi_n| dP < \varepsilon$$

for any $n = 1, \dots, N$, completing the proof.

EXERCISE 1.6. *Show that a uniformly integrable sequence of random variables is bounded in L^1 , i.e.*

$$\sup_n E(|\xi_n|) < \infty.$$

Hint: Write $E(|\xi_n|)$ as the sum of the integrals of $|\xi_n|$ over $\{|\xi_n| > M\}$ and $\{|\xi_n| \leq M\}$.

SOLUTION 1.6. *Because ξ_n is a uniformly integrable sequence, there is an $M > 0$ such that for all n*

$$\int_{\{|\xi_n| > M\}} |\xi_n| dP < 1.$$

It follows that

$$\begin{aligned} E(|\xi_n|) &= \int_{\{|\xi_n| > M\}} |\xi_n| dP + \int_{\{|\xi_n| \leq M\}} |\xi_n| dP \\ &< 1 + MP \{|\xi_n| \leq M\} \\ &< 1 + M < \infty \end{aligned}$$

for all n , proving that ξ_n is a bounded sequence in L^1 .

Exercise 1.6 implies that each uniformly integrable martingale satisfies the conditions of Doob's theorem. Therefore it converges a.s. to an integrable random variable. We shall show that in fact it converges in L^1 .

Theorem 1.9. *Every uniformly integrable supermartingale (submartingale) ξ_n converges in L^1 .*

PROOF. *By Exercise 1.6 the sequence ξ_n is bounded in L^1 , so it satisfies the conditions of Theorem 1.5 (Doob's Martingale Convergence Theorem). Therefore, there is an integrable random variable ξ such that $\xi_n \rightarrow \xi$ a.s. We can assume without loss of generality that $\xi = 0$ (since $\xi_n - \xi$ can be taken in place of ξ_n). That is to say,*

$$P \left\{ \lim_n \xi_n = 0 \right\} = 1.$$

It follows that $\xi_n \rightarrow 0$ in probability, i.e. for any $\varepsilon > 0$

$$P \{ |\xi_n| > \varepsilon \} \rightarrow 0$$

as $n \rightarrow \infty$. This is because by Fatou's lemma

$$\begin{aligned} \limsup_n P \{ |\xi_n| > \varepsilon \} &\leq P (\limsup_n \{ |\xi_n| > \varepsilon \}) \\ &\leq P (\Omega \setminus \{ \lim_n \xi_n = 0 \}) \\ &= 0. \end{aligned}$$

Let $\varepsilon > 0$. By uniform integrability there is an $M > 0$ such that

$$\int_{\{|\xi_n| > M\}} |\xi_n| dP \leq \frac{\varepsilon}{3}$$

for all n . Since $\xi_n \rightarrow 0$ in probability, there is an integer N such that if $n > N$, then

$$P \left\{ |\xi_n| > \frac{\varepsilon}{3} \right\} < \frac{\varepsilon}{3M}.$$

We can assume without loss of generality that $M > \frac{\varepsilon}{3}$. Then

$$\begin{aligned} E(|\xi_n|) &= \int_{\{|\xi_n| > M\}} |\xi_n| dP + \int_{\{M \geq |\xi_n| > \frac{\varepsilon}{3}\}} |\xi_n| dP \\ &\quad + \int_{\{\frac{\varepsilon}{3} \geq |\xi_n|\}} |\xi_n| dP \\ &\leq \frac{\varepsilon}{3} + MP\left\{|\xi_n| > \frac{\varepsilon}{3}\right\} + \frac{\varepsilon}{3}P\left\{\frac{\varepsilon}{3} \geq |\xi_n|\right\} \\ &< \varepsilon \end{aligned}$$

for all $n > N$. This proves that $E(|\xi_n|) \rightarrow 0$, that is, $\xi_n \rightarrow 0$ in L^1 . $\square$

Theorem 1.10. Let ξ_n be a uniformly integrable martingale. Then

$$\xi_n = E(\xi | \mathcal{F}_n),$$

where $\xi = \lim_n \xi_n$ is the limit of ξ_n in L^1 and $\mathcal{F}_n = \sigma(\xi_1, \dots, \xi_n)$ is the filtration generated by ξ_n .

PROOF. For any $m > n$

$$E(\xi_m | \mathcal{F}_n) = \xi_n,$$

i.e. for any $A \in \mathcal{F}_n$

$$\int_A \xi_m dP = \int_A \xi_n dP.$$

Let n be an arbitrary integer and let $A \in \mathcal{F}_n$. For any $m > n$

$$\begin{aligned} \left| \int_A (\xi_n - \xi) dP \right| &= \left| \int_A (\xi_m - \xi) dP \right| \\ &\leq \int_A |\xi_m - \xi| dP \\ &\leq E(|\xi_m - \xi|) \rightarrow 0 \end{aligned}$$

as $m \rightarrow \infty$. It follows that

$$\int_A \xi_n dP = \int_A \xi dP$$

for any $A \in \mathcal{F}_n$, so $\xi_n = E(\xi | \mathcal{F}_n)$. $\square$

EXERCISE 1.7. Show that if ξ_n is a martingale and $\xi_n \rightarrow a$ in L^1 for some $a \in \mathbb{R}$, then $\xi_n = a$ a.s. for each n .

Hint: Apply Theorem 1.10.

SOLUTION 1.7. By Theorem 1.10, $\xi_n = E(a | \mathcal{F}_n)$ a.s. But $E(a | \mathcal{F}_n) = a$ a.s., which proves that $\xi_n = a$ a.s.

Theorem 1.11 (Kolmogorov's 0–1 Law). Let $\eta_1, \eta_2, \dots$ be a sequence of independent random variables. We define the tail σ -field

$$\mathcal{T} = \mathcal{T}_1 \cap \mathcal{T}_2 \cap \dots,$$

where $\mathcal{T}_n = \sigma(\eta_n, \eta_{n+1}, \dots)$. Then

$$P(A) = 0 \text{ or } 1$$

for any $A \in \mathcal{T}$.

PROOF. Take any $A \in \mathcal{T}$ and define

$$\xi_n = E(1_A | \mathcal{F}_n),$$

where $\mathcal{F}_n = \sigma(\eta_1, \dots, \eta_n)$. By Exercise 1.5 ξ_n is a uniformly integrable martingale, so $\xi_n \rightarrow \xi$ in L^1 . By Theorem 1.10

$$E(\xi | \mathcal{F}_n) = E(1_A | \mathcal{F}_n)$$

for all n . Both $\xi = \lim_n \xi_n$ and 1_A are measurable with respect to the σ -field

$$\mathcal{F}_\infty = \sigma(\eta_1, \eta_2, \dots).$$

The family $\mathcal{G}$ consisting of all sets $B \in \mathcal{F}$ such that $\int_B \xi dP = \int_B 1_A dP$ is a σ -field containing $\mathcal{F}_1 \cup \mathcal{F}_2 \cup \dots$. As a result, $\mathcal{G}$ contains the σ -field $\mathcal{F}_\infty$ generated by the family $\mathcal{F}_1 \cup \mathcal{F}_2 \cup \dots$. By Lemma 2.1 it follows that $\xi = 1_A$ a.s.

Since η_n is a sequence of independent random variables, the σ -fields $\mathcal{F}_n$ and $\mathcal{T}_{n+1}$ are independent. Because $\mathcal{T} \subset \mathcal{T}_{n+1}$, the σ -fields $\mathcal{F}_n$ and $\mathcal{T}$ are independent. Being $\mathcal{T}$ -measurable, 1_A is therefore independent of $\mathcal{F}_n$ for any n . This means that

$$\xi_n = E(1_A | \mathcal{F}_n) = E(1_A) = P(A) \quad \text{a.s.}$$

Therefore the limit $\lim_{n \rightarrow \infty} \xi_n = \xi$ is also constant and equal to $P(A)$ a.s. This means that $P(A) = 1_A$ a.s., so $P(A) = 0$ or 1 . $\square$

EXERCISE 1.8. Show that if $A_n \in \sigma(\xi_n)$ for each n , then the events

$$\limsup_n A_n = \bigcap_{j \geq 1} \bigcup_{i \geq j} A_i$$

and

$$\liminf_n A_n = \bigcup_{j \geq 1} \bigcap_{i \geq j} A_i$$

belong to the tail σ -field $\mathcal{T}$.

Hint: You need to write $\limsup_n A_n$ and $\liminf_n A_n$ in terms of $A_k, A_{k+1}, \dots$ for any k , that is, to show that $\limsup_n A_n$ and $\liminf_n A_n$ will not be affected if any finite number of sets are removed from the sequence $A_1, A_2, \dots$

SOLUTION 1.8. Observe that

$$\limsup_n A_n = \bigcap_{j \geq k} \bigcup_{i \geq j} A_i$$

for any k . Since

$$A_i \in \sigma(\xi_i) \subset \mathcal{T}_k$$

for every $i \geq k$, it follows that

$$\limsup_n A_n = \bigcap_{j \geq k} \bigcup_{i \geq j} A_i \in \mathcal{T}_k$$

for every k . Therefore

$$\sup_n A_n \in \mathcal{T}.$$

The argument for $\liminf_n A_n$ is similar.

EXERCISE 1.9. Use Kolmogorov's 0–1 law to show that in a sequence of coin tosses there are a.s. infinitely many heads.

Hint: Show that the event

$$\{\eta_1, \eta_2, \dots \text{ contains infinitely many heads} \}$$

belongs to the σ -field $\mathcal{T}$. Can the probability of this event be 0? The probability of the event

$$\{\eta_1, \eta_2, \dots \text{ contains infinitely many tails} \}$$

should be the same. Can both probabilities be equal to 0? Can you simultaneously have finitely many heads and finitely many tails in the sequence $\eta_1, \eta_2, \dots$?

SOLUTION 1.9. Let $\eta_1, \eta_2, \dots$ be a sequence of coin tosses, i.e. independent random variables with values $+1, -1$ (heads or tails) taken with probability $\frac{1}{2}$ each. Consider the following event:

$$A = \{ \eta_1, \eta_2, \dots \text{ contains infinitely many heads} \}.$$

This event belongs to the tail σ -field $\mathcal{T}$ because

$$A = \limsup_n A_n,$$

where

$$A_n = \{ \eta_n = \text{heads} \} \in \sigma(\eta_n)$$

(see Exercise 1.8). Thus, by Kolmogorov's 0-1 law $P(A) = 0$ or 1 . However, it cannot be 0 because the event

$$B = \{ \eta_1, \eta_2, \dots \text{ contains infinitely many tails} \}$$

has the same probability by symmetry and $\Omega = A \cup B$ (there must be infinitely many heads or infinitely many tails).

CHAPTER 2

Markov Chains

1. First Examples and Definitions

1.1. Example 5.1. In some homes the use of the telephone can become quite a sensitive issue. Suppose that if the phone is free during some period of time, say the n th minute, then with probability p , where $0 < p < 1$, it will be busy during the next minute. If the phone has been busy during the n th minute, it will become free during the next minute with probability q , where $0 < q < 1$. Assume that the phone is free in the 0th minute. We would like to answer the following two questions.

- (1) What is the probability x_n that the telephone will be free in the n th minute?
- (2) What is $\lim_{n \rightarrow \infty} x_n$, if it exists?

Denote by A_n the event that the phone is free during the n th minute and let $B_n = \Omega \setminus A_n$ be its complement, i.e. the event that the phone is busy during the n th minute. The conditions of the example give us

$$(5.1) \quad P(B_{n+1}|A_n) = p,$$

$$(5.2) \quad P(A_{n+1}|B_n) = q.$$

We also assume that $P(A_0) = 1$, i.e. $x_0 = 1$. Using this notation, we have $x_n = P(A_n)$. Then the total probability formula, see Exercise 1.10, together with (5.1)-(5.2) imply that

$$(5.3) \quad x_{n+1} = P(A_{n+1})$$

$$(1) \quad = P(A_{n+1}|A_n)P(A_n) + P(A_{n+1}|B_n)P(B_n)$$

$$(2) \quad = (1-p)x_n + q(1-x_n) = q + (1-p-q)x_n.$$

It's a bit tricky to find an explicit formula for x_n . To do so we suppose first that the sequence $\{x_n\}$ is convergent, i.e.

$$(5.4) \quad \lim_{n \rightarrow \infty} x_n = x.$$

The elementary properties of limits and equation (5.3), i.e. $x_{n+1} = q + (1-p-q)x_n$, yield

$$(5.5) \quad x = q + (1-p-q)x.$$

The unique solution to the last equation is

$$(5.6) \quad x = \frac{q}{q+p}.$$

In particular,

$$(5.7) \quad \frac{q}{q+p} = q + (1-p-q) \frac{q}{q+p}.$$

Subtracting (5.7) from (5.3), we infer that

$$(5.8) \quad x_{n+1} - \frac{q}{q+p} = (1-p-q) \left(x_n - \frac{q}{q+p} \right).$$

Thus, $\{x_n - \frac{q}{q+p}\}$ is a geometric sequence and therefore, for all $n \in \mathbb{N}$,

$$(3) \quad x_n - \frac{q}{q+p} = (1-p-q)^n \left(x_0 - \frac{q}{q+p} \right).$$

Hence, by taking into account the initial condition $x_0 = 1$, we have

$$(5.9) \quad x_n = \frac{q}{q+p} + \left(x_0 - \frac{q}{q+p} \right) (1-p-q)^n$$

$$(4) \quad = \frac{q}{q+p} + \frac{p}{q+p} (1-p-q)^n.$$

Let us point out that although we have used the assumption (5.4) to derive (5.8), the proof of the latter is now complete. Indeed, having proven (5.9), we can show that the assumption (5.4) is indeed satisfied. This is because the conditions $0 < p, q < 1$ imply that $|1-p-q| < 1$, and so $(1-p-q)^n \rightarrow 0$ as $n \rightarrow \infty$. Thus, (5.4) holds. This provides an answer to the second part of the example, i.e. $\lim_{n \rightarrow \infty} x_n = \frac{q}{p+q}$.

EXERCISE 2.1. *In the framework of Example 1.1, let y_n denote the probability that the telephone is busy in the n th minute. Supposing that $y_0 = 1$, find an explicit formula for y_n and, if it exists, $\lim_{n \rightarrow \infty} y_n$.*

REMARK 2.1. ***Hint:** This exercise can be solved directly by repeating the above argument, or indirectly by using some of the results in Example 1.1.*

SOLUTION 2.1. *Since the telephone must be either free or busy at any time n , we have $x_n + y_n = 1$, where x_n is the probability that the telephone is free. If $y_0 = 1$, then $x_0 = 0$. Substituting $x_0 = 0$ into the formula (5.9) for x_n derived in Example 1.1, we obtain*

$$x_n = \frac{q}{q+p} + \left(0 - \frac{q}{q+p} \right) (1-p-q)^n = \frac{q}{q+p} - \frac{q}{q+p} (1-p-q)^n.$$

Therefore,

$$y_n = 1 - x_n = 1 - \frac{q}{q+p} + \frac{q}{q+p} (1-p-q)^n = \frac{p}{q+p} + \frac{q}{q+p} (1-p-q)^n.$$

As $n \rightarrow \infty$, since $|1-p-q| < 1$, the term $(1-p-q)^n$ tends to zero, so

$$\lim_{n \rightarrow \infty} y_n = \frac{p}{q+p}.$$

REMARK 2.2. The formulae (5.3) and (5.9) can be written collectively in a compact form by using vector and matrix notation. First of all, since $x_n + y_n = 1$, we get

$$(5) \quad x_{n+1} = (1 - p)x_n + qy_n,$$

$$(6) \quad y_{n+1} = px_n + (1 - q)y_n.$$

Hence, the matrix version takes the form

$$(7) \quad \begin{bmatrix} x_{n+1} \\ y_{n+1} \end{bmatrix} = \begin{bmatrix} 1 - p & q \\ p & 1 - q \end{bmatrix} \begin{bmatrix} x_n \\ y_n \end{bmatrix}.$$

The situation described in Example 1.1 is quite typical. Often the probability of a certain event at time $n + 1$ depends only on what happens at time n , but not further into the past. Example 1.1 provides us with a simple case of a Markov chain. See also the following definition and exercises.

DEFINITION 2.1 (Markov chain). Suppose that S is a finite or a countable set. Suppose also that a probability space $(\Omega, \mathcal{F}, P)$ is given. An S -valued sequence of random variables ξ_n , $n \in \mathbb{N}$, is called an S -valued Markov chain or a Markov chain on S if for all $n \in \mathbb{N}$ and all $s \in S$

$$(5.10) \quad P(\xi_{n+1} = s \mid \xi_0, \dots, \xi_n) = P(\xi_{n+1} = s \mid \xi_n).$$

Here $P(\xi_{n+1} = s \mid \xi_n)$ is the conditional probability of the event $\{\xi_{n+1} = s\}$ with respect to random variable ξ_n , or equivalently, with respect to the σ -field $\sigma(\xi_n)$ generated by ξ_n . Similarly, $P(\xi_{n+1} = s \mid \xi_0, \dots, \xi_n)$ is the conditional probability of $\{\xi_{n+1} = s\}$ with respect to the σ -field $\sigma(\xi_0, \dots, \xi_n)$ generated by the random variables $\xi_0, \dots, \xi_n$.

Property (5.10) will usually be referred to as the Markov property of the Markov chain $\xi_n, n \in \mathbb{N}$. The set S is called the state space and the elements of S are called states.

Proposition 2.2. The model in Example 1.1 and Exercise 2.1 is a Markov chain.

PROOF. Let $S = \{0, 1\}$, where 0 and 1 represent the states of the phone being free or busy. First we need to construct an appropriate probability space. Let Ω be the set of all sequences $\omega_0, \omega_1, \dots$ with values in S . Let μ_0 be any probability measure on S . For example, $\mu = \delta_0$ corresponds to the case when the phone is free at time 0. We shall define P by induction. For any S -valued sequence $s_0, s_1, \dots$ we put

$$(5.11) \quad P(\{\omega \in \Omega : \omega_0 = s_0\}) = \mu_0(\{s_0\})$$

and

$$(5.12) \quad P(\{\omega \in \Omega : \omega_i = s_i, i = 0, \dots, n + 1\})$$

$$(8) \quad = p(s_{n+1} \mid s_n)P(\{\omega \in \Omega : \omega_i = s_i, i = 0, \dots, n\}),$$

where $p(s \mid r)$ are the entries of the 2×2 matrix

$$(9) \quad \begin{bmatrix} p(0 | 0) & p(0 | 1) \\ p(1 | 0) & p(1 | 1) \end{bmatrix} = \begin{bmatrix} 1-p & q \\ p & 1-q \end{bmatrix}.$$

It seems reasonable to expect P to be a probability measure (with respect to the trivial σ -field of all subsets of Ω). Take this for granted, check only that $P(\Omega) = 1$.

How would you define the process $\xi_n, n \in \mathbb{N}$? We shall do it in the standard way, i.e.

$$(5.13) \quad \xi_n(\omega) = \omega_n, \omega \in \Omega.$$

First we shall show that the transition probabilities of ξ_n are what they should be, i.e.

$$(5.14) \quad P(\xi_{n+1} = 1 \mid \xi_n = 0) = p,$$

$$(5.15) \quad P(\xi_{n+1} = 0 \mid \xi_n = 1) = q.$$

The definition of conditional probability yields

$$(10) \quad P(\xi_{n+1} = 1 \mid \xi_n = 0) = \frac{P(\xi_{n+1} = 1, \xi_n = 0)}{P(\xi_n = 0)}.$$

Next, the definition of P gives

$$(11) \quad \begin{aligned} P(\xi_{n+1} = 1, \xi_n = 0) &= P(\{\omega \in \Omega : \omega_n = 0, \omega_{n+1} = 1\}) \\ &= \sum_{s_0, \dots, s_{n-1} \in S} P(\{\omega \in \Omega : \omega_i = s_i, i = 0, \dots, n-1, \omega_n = 0, \omega_{n+1} = 1\}) \\ &= \sum_{s_0, \dots, s_{n-1} \in S} pP(\{\omega \in \Omega : \omega_i = s_i, i = 0, \dots, n-1, \omega_n = 0\}) \\ &= pP(\xi_n = 0), \end{aligned}$$

by (5.12). We have proven (5.14). Moreover, (5.15) follows by the same argument. A similar line of reasoning shows that ξ_n is indeed a Markov chain. For this we need to verify that for any $n \in \mathbb{N}$ and any $s_0, s_1, \dots, s_{n+1} \in S$

$$(12) \quad P(\xi_{n+1} = s_{n+1} \mid \xi_0 = s_0, \dots, \xi_n = s_n) = P(\xi_{n+1} = s_{n+1} \mid \xi_n = s_n).$$

We have

$$(13) \quad \begin{aligned} &P(\xi_0 = s_0, \dots, \xi_n = s_n, \xi_{n+1} = s_{n+1}) \\ &= P(\{\omega \in \Omega : \omega_i = s_i, i = 0, \dots, n+1\}) \\ &= p(s_{n+1} \mid s_n)P(\{\omega \in \Omega : \omega_i = s_i, i = 0, \dots, n\}) \\ &= p(s_{n+1} \mid s_n)P(\xi_0 = s_0, \dots, \xi_n = s_n) \end{aligned}$$

which, in view of the definition of conditional probability, gives